

Part 1a: Navigating public databases and resources

The purpose of Part 1 is to learn how to navigate and use public databases and resources to understand the inheritance pattern, genomic architecture, and pathogenesis pathway of genetic diseases.

IMPORTANT:

- Your group is assigned to a gene/variant of interest (**G/VOI**) from a genetic disease (that you have selected as a group).
- Complete and submit Lab 1 individually using this gene/variant of interest and disease.
- You may work together as a group to verify the information you've collected across the databases.

Here is a table of the resources we will be using in Lab 1. This is a non-comprehensive table of publicly available resources, but we will use the following to get started.

Database	Purpose
UCSC Genome Browser	Provides genomic and epigenomic data, including visualization tools for genome annotations.
OMIM	A database of human genes and genetic disorders. Use this to investigate gene-disease relationships and find information on genetic disorders.
GeneCards	Integrates information about genes, their functions, expression, and and relevance to diseases.
GeneReviews	Expert-authored summaries on genetic conditions, including diagnostic and treatment guidelines.

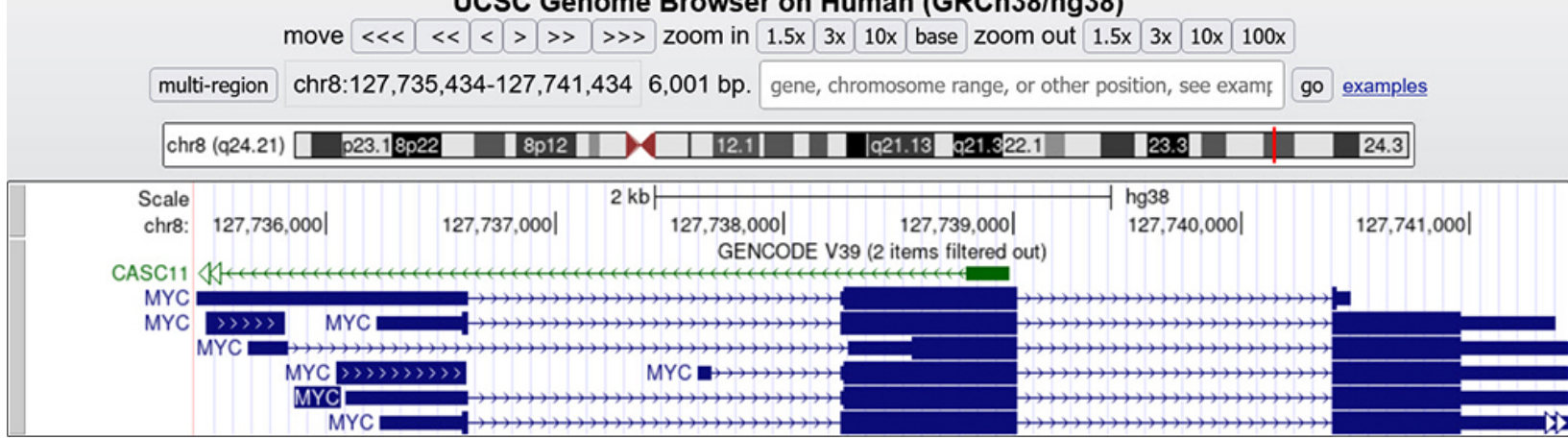
Exercise 1: UCSC Genome Browser

Genome browsers *integrate genomic sequence* and *annotation data* from different sources and provide a platform to search, browse, retrieve, and analyze genomic data. They differ from ordinary biological databases in that they display data in a graphical format, with genome coordinates on one axis with annotations or space-filling graphics to show analyses of the genes, such as the frequency of the genes, their expression profiles, etc.

1a: Find your G/VOI (1.5 pt)

Instructions: Complete the following using your **G/VOI**

- Go to the **UCSC Genome Browser**, center the track on your gene or variant of interest. If you are assigned a genetic variant, include the entire gene that the variant belongs to.
- Locate the track showing the exons, introns, and direction of strands like the image below.
 - The direction of arrows indicates the 5' > 3' direction of a gene.
 - Each row of the blue genes is an isoform.



In the code cell below:

- (0.5 pt) Store the location of the gene (transcript including UTR's) as shown in UCSC Genome Browser as a string in the variable `gene_location`.
- (0.5 pt) Store the strand direction as a string in the variable `strand_direction`.
- (0.5 pt) Store the number of exons as an integer in the variable `total_exon`.

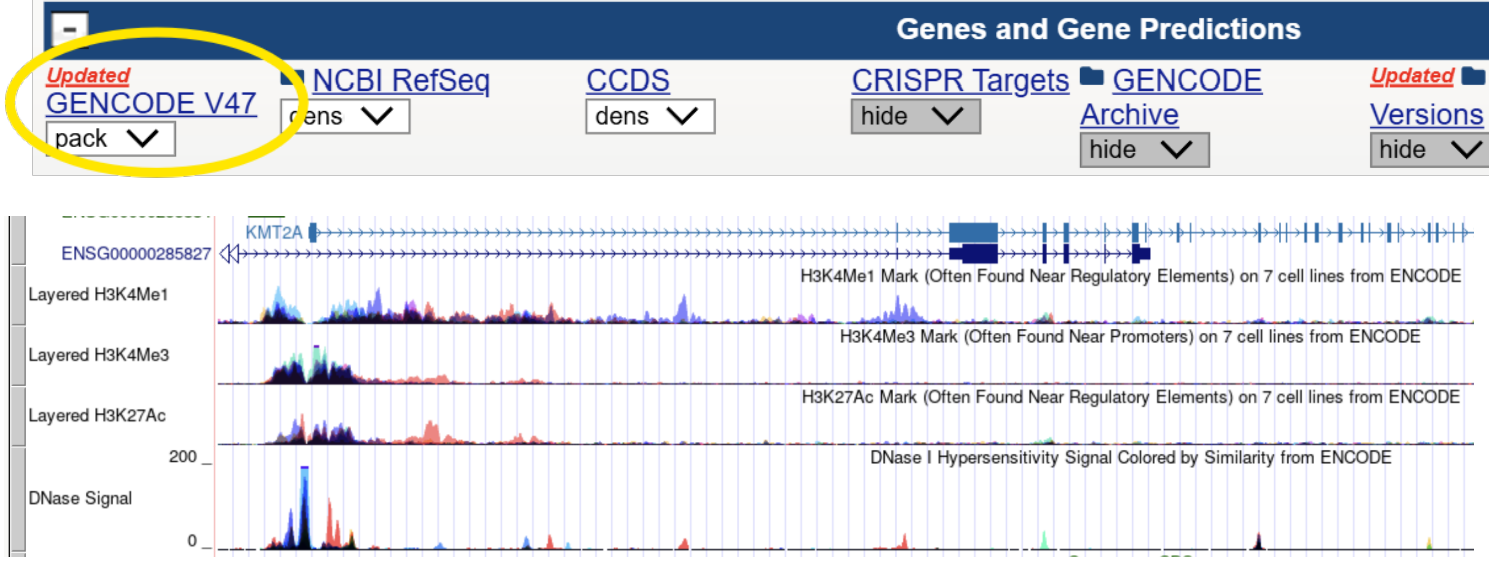
Here is an example for the gene KMT2A:

```
gene_location = "chr11:118436492-118526832"
strand_direction = "+"
total_exon = 36
```

In [3]: `# Enter your answer here.`

1b: Locate the regulatory elements using ENCODE tracks (2 pt)

- Add the **ENCODE track** to your UCSC Genome Browser.



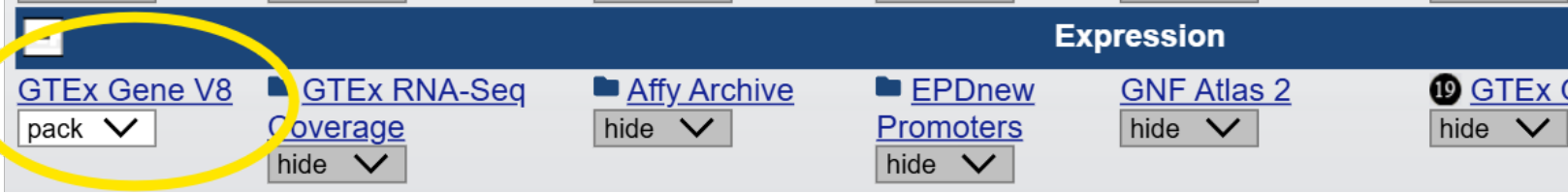
- (2 pt) In 2-3 sentences, explain the locations of the peaks in the H3K Tracks and DNase I Hypersensitivity Track. Based on your knowledge of the gene structure, why do you see the peaks at these locations?

YOUR ANSWER HERE

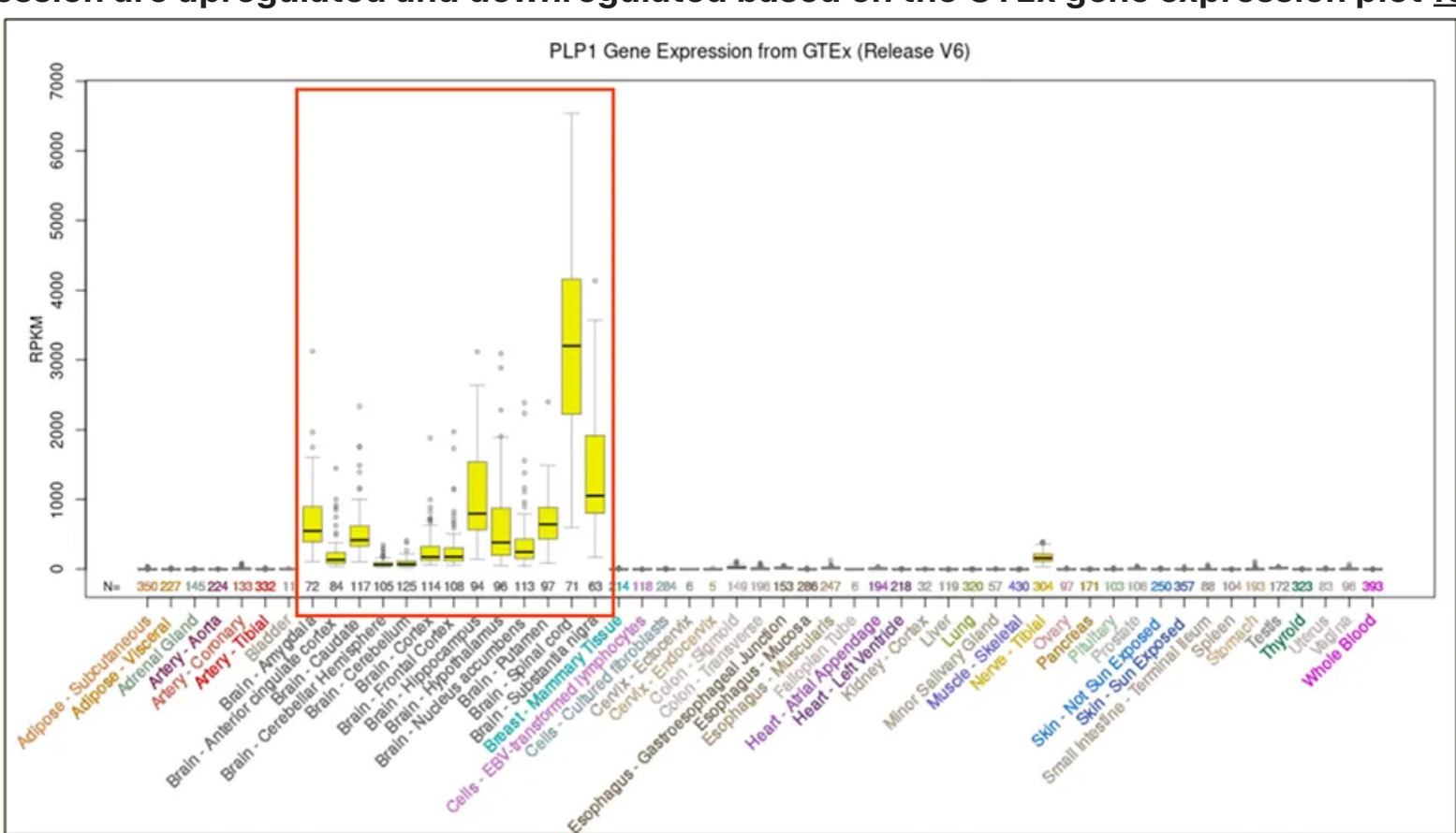
1c: Find tissue-specific gene expression using GTEx V8 track (2 pt)

Instructions:

- Add the **GTEx Gene v8 track** to your UCSC Genome Browser.
- Click on the added track to find the tissue-specific gene expression.
- In two sentences, describe the tissues in which the gene expression are upregulated and downregulated based on the GTEx gene expression plot for your assigned G/VOI.



Your plot should have the same x- and y-axes as the figure below.

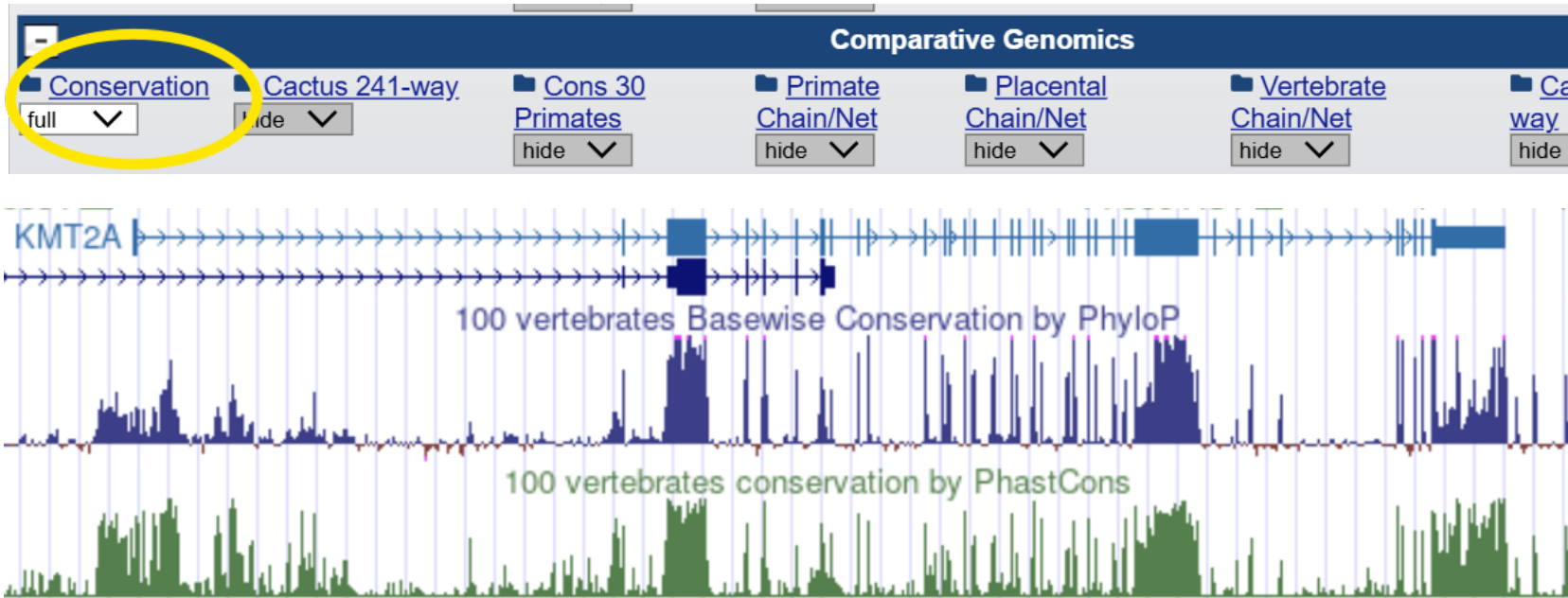


YOUR ANSWER HERE

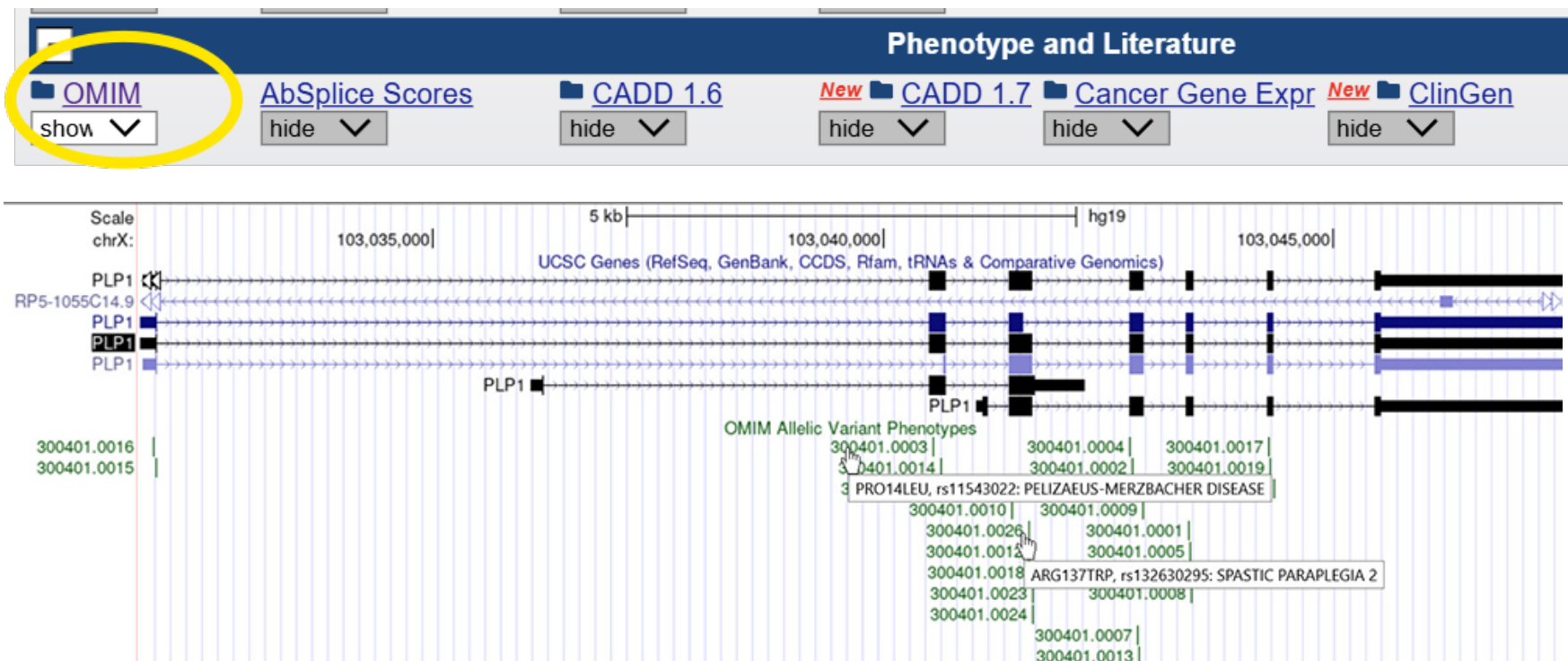
1d: Find conserved regions using conservation tracks (2 pt)

Instructions:

- Add the **Conservation track** to your UCSC Genome Browser



- Add the **OMIM track** to locate the disease variants.



Provide the answer using your G/VOI in the markdown cell below:

- (0.5 pt) List one disease variant that lies in a region with high conservation score.
- (0.5 pt) List one disease variant that is in a region with low conservation score.
- (1 pt) What does the conservation score suggest about the rate of evolution for these disease variant

YOUR ANSWER HERE

1e: Screenshot of your UCSC Genome Browser (1 pt)

Instructions:

- Attach a screenshot of your UCSC Genome Browser in the markdown cell below.
- Include all of the tracks mentioned in Exercise 1.

YOUR ANSWER HERE

Exercise 2: OMIM Database

Instructions:

- Go to **OMIM** and look up your **G/VOI**.

OMIM®

An Online Catalog of Human Genes and Genetic Disorders

Updated November 15th, 2024

- Select "Gene Map Table"

View Results
Display: ☒ Highlight

Search: 'KMT2A'
Results: 15 entries.

Show 100 |

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- Click on the **Gene/Locus MIM** and **Phenotype MIM** numbers to find clinically relevant information about your **G/VOI**.

Search: 'KMT2A' (Search in: Entries with: Genemap; Retrieve: gene map)
Results: 13 entries.

Show 100 |

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Phenotype Only Entries | All Entries

Genomic context table	Location (from NCBI, GRCh38)	Gene/Locus	Gene/Locus name	Gene/Locus MIM number	Phenotype	Phenotype MIM number	Inheritance	Pheno map key	Comments	Mouse symbol (from MGI)
1:	2:218,710,835-2q35	TTL4	Tubulin tyrosine ligase-like 4	618738						Ttl4
2:	4:48,497,357-4p11	FRYL, KIAA0826	FRY-like transcription coactivator	620798						Fryl
3:	8:41,929,479-8p11.21	KAT6A, MYST3, MOZ, ZNF220, ARTHS	K(lysine) acetyltransferase 6A	601408	Arboleda-Tham syndrome	616268	AD	3		Kat6a
4:	10:806,914-10p15.3	LARP4B, LARP5, KIAA0217	La ribonucleoprotein 4B	616513						Larp4b
5:	11:118,436,492-11q23.3	KMT2A, MLL, HRX, WDST5	Lysine-specific methyltransferase 2E	159555	Wiedemann-Steiner syndrome	605130	AD	3	fuses with EN1, AF4, AF9, GMPS	Kmt2a

Provide the answer using your G/VOI in the markdown cell below:

- (0.5 pt) What is the inheritance pattern of your disease?
- (0.5 pt) What is the most common type of allelic variant associated with your GVOI?

YOUR ANSWER HERE

Exercise 3: Pathogenesis pathway (2 pt)

Based on the information in the first two exercises and accessible information in GeneCards and GeneReviews, describe the pathogenesis pathway of your assigned disease. Specifically, explain how your GVOI may contribute to the clinical phenotype of the disease. Cite the database(s) from which you found the information.

YOUR ANSWER HERE